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Hybrid image encryption based on Hessenberg decomposition and chaotic structured phase masks in Fresnel transform domain

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Abstract
This system uses the Hessenberg decomposition (HD) and Chaotic structured phase masks (CSPMs) to propose a phase image encryption algorithm. HD makes the cryptosystem asymmetric. The use of CSPMs enhances the security of the proposed cryptosystem. The approach is further secured by the parameters of the Fresnel transform (FrT). The effectiveness of this method is examined using noise and occlusion attacks. To bolster its legitimacy, key sensitivity results were also included. The obtained values of mean squared error (MSE) is 1.68 $\times 10^{-31}$ for grey image and 1.62 $\times 10^{-31}$ for binary image which represents the effectiveness of the proposed cryptosystem. Parameters of Fresnel transform ($z_1 = 25$ mm, $z_2 = 35$ mm, $\lambda = 632.8$ nm) are used to enhance the security of the proposed scheme. This outcome exhibits that the method is effective in obtaining safe image encryption, providing excellent image reconstruction with minimal information loss.

Keywords Chaotic structured phase masks · Hessenberg decomposition · Fresnel transform · Key sensitivity analysis

1 Introduction

With the advancement of technology and the internet, communication has become a necessity in people's lives. Due to its inherent benefits of large degrees of freedom and high speed parallel 2D data processing, optical information security has been extensively studied. Social media data sharing-text, image, and video- has grown to be a significant aspect of people's

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everyday lives. Unauthorized data leaks can be detrimental to people, companies, and occasionally the entire countries. Numerous encryption methods exist (Singh, 2013), and they are crucial in preventing unauthorized individuals from accessing data. Digital image encryption methods are more vulnerable and slower when calculations are involved. Double random phase encoding (DRPE) is the name of an optical encryption technique that Refregier and Javidi (Refregier & Javidi, 1995) presented in 1995. Due to its optical properties-such as fast speed, parallelism, and multi-dimensional key space-many researchers began to focus on DRPE. In DRPE, two random phase masks (RPMs) are used; one in the frequency domain and the other in the spatial domain. DRPE schemes are built in different transform domains, including Hartley transform (Chen & Zhao, 2006), fractional Hartley transform (Singh, 2017; Singh et al., 2017), fractional Mellin transform (Vashisth et al., 2014; Zhou & Gong, 2011), fractional Fourier transform (Girija & Singh, 2018a; Hsue, 2018; Singh & Sinha, 2008; Unnikrishnan et al., 2000), Gyrator transform (Rodrigo et al., 2007; Singh et al., 2014; Yadav et al., 2015), Fresnel transform (Situ & Zhang, 2004; Situ et al., 2010), and so on. The DRPE technique, which is essentially a symmetric cryptosystem, uses the same keys for both encryption and decryption, leading to problems with key distribution. DRPE is vulnerable to several attacks (Carnicer et al., 2005; Gabr et al., 2023; Peng et al., 2006a, 2006b), including known-plaintext attacks (KPA), chosen plaintext attacks (CPA), and chosen ciphertext attacks (CCA), due to its linear and symmetric structure. As a result, scientists began to concentrate on asymmetric cryptosystems for DRPE systems. Qin and Peng (Qin & Peng, 2010) proposed the first asymmetric cryptosystem using the phase truncation Fourier transform (PTFT). Two RPMs serve as the public key in the PTFT cryptosystem, while the private key is derived from phase reservation operations. The phase truncation technique yielded a genuine value for the encrypted image. Later, it was discovered that the phase retrieval algorithm based PTFT cryptosystem is vulnerable to a specific attack (Wang & Zhao, 2012), where the attacker can obtain plaintext without requiring decryption keys. For added security, the phase-truncation-based encryption method has also been expanded into the Fresnel or fractional Fourier domains. Additionally, the authors proposed a cryptosystem that retrieves the phase and amplitude combination of the Yang-Gu algorithm. A straightforward optical cryptosystem that employs a random amplitude mask (RAM), directly attack PTFT-based encryption (Liansheng & Zhanmin, 2016), with no iterative computations required for the nonlinear encryption. Polar decomposition (Gaur et al., 2024; Oussama et al., 2017), lower upper decomposition (LUPD) (Anshula, 2021; Muhammad & Bibi, 2015; Wang et al., 2021; Xiong & Quan, 2019), equal modulus decomposition (EMD) (Cai et al., 2015; Chen et al., 2017, 2019; Fatima et al., 2016; Shao et al., 2020; Wang et al., 2019), and Hessenberg decomposition (HD) (Su & Chen, 2017; Wang & Ji, 2022) are among the decomposition classes that have been used in asymmetric encryption. In the place of traditional phase masks in DRPE, CSPMs are produced based on the logistic map, Fresnel zone plates (FZP) (Barrera et al., 2005a, 2005b), and radial Hilbert mask (RHM) (Girija & Singh, 2018b) as random masks to increase the security and irregularity in DRPE. Hence, our proposed hybrid cryptosystem generates the CSPM in the place of traditional masks and uses the Hessenberg Decomposition.

The structure of this document is as follows: Sect. 1 presents introduction; The mathematical underpinnings of the CSPMs, HD and Fresnel transform are thoroughly examined in Sect. 2. Our suggested method for proposed image encryption is explained in Sect. 3. Numerical simulations demonstrating the viability and efficiency of the suggested approach are presented in Sect. 4. Section 5 concludes with some final thoughts.

2 Mathematical background

The background for the proposed methodology includes CSPMs, FrT, HD. A detailed description of each is provided below.

2.1 Chaotic structured phase masks (Alexan et al., 2023a, 2023b, 2023c, 2024; El-Damak et al., 2024)

Random phase masks (RPM) are used as secret keys in DRPE for image encryption. These RPMs are also constructed using different chaotic maps. The chaos function represents the transition of nonlinear systems from an organised state to a chaotic state. The properties of most random iterative values generated by a chaotic map are pseudo-randomness, ergodicity, and non-correlation. One such chaotic function is the logistic map, a one-dimensional (1D) nonlinear function, defined as,

$$f(x) = p \cdot x \cdot (1 - x) \tag{1}$$

where p as the bifurcation factor. The formula is iterated and provided as,

$$x_{n+1} = p \cdot x_n \cdot (1 - x_n) \tag{2}$$

where $x_n \in (0, 1)$ and x_0 is the iterated value and first value respectively. Equation (2) requires p and x_0 as seed values because it is a recursive function. Even a tiny alteration in x_0 and $p \in \{3.5699456, 4\}$, causes the sequence x_n to become chaotic. The resulting chaotic sequences are non-periodic and do not converge over time.

Consider a plain image with M × N pixels. The construction of CSPMs involves three processes (a) radial Hilbert (RH) (b) Fresnel zone plates (FZP), and (c) chaotic random phase mask (CRPM). The one-dimensional random sequence, or CRPM, is generated by using Eq. (3) as

$$X = \{x_1, x_2, \ldots\ldots x_{M\times N}\}; x_i \in (0, 1) \tag{3}$$

Equation (4) represents the rearranged sequence X as Y, which is created as a two-dimensional matrix.

$$Y = \{y_{i,j} | i = 1, 2, \ldots . M; j = 1, 2, \ldots . N\}; y_{i,j} \in \{0, 1\} \tag{4}$$

CRPM sequences are currently obtained as,

$$CRPM(x, y) = \exp\left(i 2\pi y_{i,j}\right) \tag{5}$$

where (x, y) are the CRPM coordinates. Although using chaotic RPMs simplifies managing private keys easier, it significantly impairs the alignment of optical setup axis. SPM-based encryption techniques such as FZP and RHM have been proposed as solutions to this issue. The SPM incorporates features from several methods in a single mask; these factors are typically used as additional keys to enhance cryptosystem's security. The complex field amplitude distribution of FZP, as a diffractive optical element, can be expressed as,

$$FZP(r) = \exp\left(-\frac{i\pi}{\lambda f} r^2\right) \tag{6}$$

where f, and λ stand for the radius, focal length, and illuminating wavelength, respectively. The z-direction and the optical axis should line up. Construction CSPMs are shown in Fig. 1. Figure 1a shows the FZP, Fig. 1b and c shows the Radial Hilbert *for* $m = 5$ *and* $m = 7$

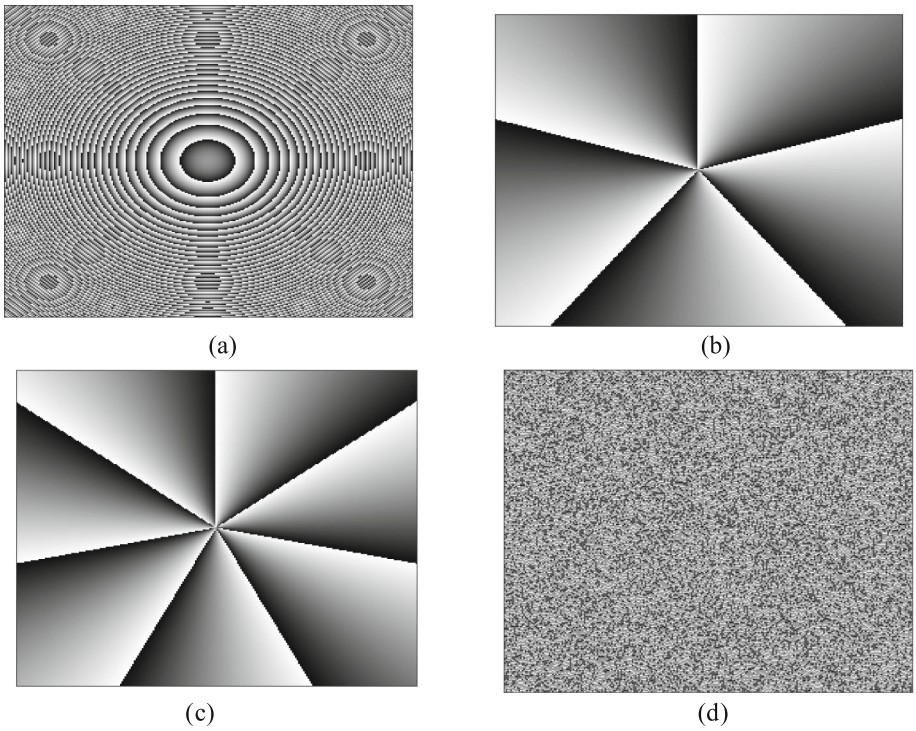


Fig. 1 CSPMs construction **a** Fresnel zone plate **b** Radial Hilbert for $m = 5$; **c** Radial Hilbert for $m = 7$ **d** Chaotic random phase masks

respectively. Figure 1d Construction of chaotic random phase masks Fig. 2a and b represents the $CSPM_1$ and $CSPM_2$.

To enhance the edges of an image, the radial Hilbert function can be defined as follows:

$$RHT(r, \theta) = \exp(im\theta) \tag{7}$$

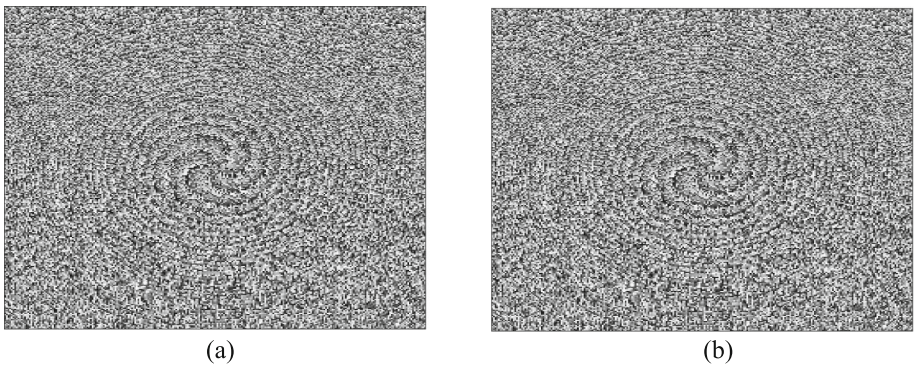


Fig. 2 CSPMs construction **a** $CSPM_1$ **b** $CSPM_2$

where r, θ are log polar coordinates; $r^2 = x^2 + y^2$ and $\theta = \tan^{-1}\left(\frac{y}{x}\right)$. The order of transformation known as topological charge is denoted by m.

To improve the security of image encryption systems, SPM can also be formed by multiplying FZP and RHT. The parameters f, λ of FZP and m of RHT are meant as private keys. SPM is stated as

$$SPM = FZP(r) \times RHT(r, \theta) \tag{8}$$

CSPMs is finally built as follows,

$$\begin{aligned} CSPM_1(x, y) &= \exp\{i\{\arg\{CRPM(x, y)\} \times \arg\{FZP(r)\} \times \arg\{RHT(r, \theta)\}\}\} \\ CSPM_2(x, y) &= \exp\{i\{\arg\{CRPM(x, y)\} \times \arg\{FZP(r)\} \times \arg\{RHT(r, \theta)\}\}\} \end{aligned} \tag{9}$$

where phase action is represented by $arg\{.\}$. For constructing $CSPM_1$ and $CSPM_2$ value of m is 5 and 7 respectively.

2.2 Fresnel transform (Situ & Zhang, 2004; Situ et al., 2010)

When an input image $f(x, y)$ is illuminated by a plane wave of wavelength λ, its Fresnel transform (FrT) at a propagation distance z can be expressed as,

$$F_z(u, v) = FrT_{\lambda, z}\{f(x, y)\} = \int_{-\infty}^{+\infty} \int f(x, y) h_{\lambda, z}(u, v, x, y) dx \cdot dy \tag{10}$$

where the Fresnel transform with parameters λ and z is indicated by the operator $FrT_{\lambda, z}$, and the kernel of the transform provided by,

$$h_{\lambda, z}(u, v, x, y) = \frac{1}{\sqrt{i\lambda z}} \exp\left(i \frac{2\pi z}{\lambda}\right) \exp\left\{\frac{i\pi}{\lambda z}(u - x)^2 + (v - y)^2\right\} \tag{11}$$

A useful property of the FrT is.

$$FrT_{\lambda, z_1}\left\{FrT_{\lambda, z_2} f(x)\right\} = FrT_{\lambda, z_1 + z_2}\{f(x)\} \tag{12}$$

To satisfy the Fresnel approximation, the distance parameters z_1 and z_2 are selected based on aperture size. A Fresnel transform is then used to determine the complex amplitude distributions in the adjacent planes with respect to z_1, z_2 and λ.

2.3 Hessenberg decomposition(Shao et al., 2020; Su & Chen, 2017)

Matrix decomposition is the process by which an algorithm, based on certain principles, divides a matrix into the product of several matrices. There are various types of matrix decompositions exist, including HD and SVD. In our proposed model, HD is used to enhance the robustness and stability of our system. HD is employed to split the images and is defined as a special square matrix, which is given below:

$$I = PHP^T \tag{13}$$

where I is the overall index that matches the image. Orthogonal matrix and upper Hessenberg matrix are denoted by P and H respectively.

$$\begin{cases} H = \left(h_{ij}\right)_{n \times n} \in R^{n \times n} \\ h_{ij} = 0 (j > i + 1) \end{cases} \tag{14}$$

3 Proposed model

The encryption process is given below in Fig. 3a.

Step 1: Assume that $I(x, y)$ is the input image. Figure 1 illustrates how the initial chaotic structured phase mask (CSPM1) is created before applying the Fresnel transform. The m value of $CSPM_1$ is 5.

Step 2: The input image is first converted into phase image, which is then multiplied by $CSPM_1$. Afterward, a Fresnel transform is performed to obtain the intermediate image $T(u, v)$. The description of this step is given in Eq. (15)

$$T(u, v) = FrT_{z_1}\left(\left(e^{i\pi I(x,y)}\right) * CSPM_1\right) \tag{15}$$

Step 3: HD is applied on $T(u, v)$, key is taken out of the process. Extraction of Key1 is given in below equations.

$$HD(T(u, v)) = [P_2, H_2] \tag{16}$$

$$key1 = P_2 \tag{17}$$

Step 4: Fig. 2 illustrates the second chaotic structured phase mask ($CSPM_2$). To get the encrypted image $E(x, y)$, the resultant of H_2] is multiplied with CSPM2 and it has been given to Fresnel transform (Z2). The m value of $CSPM_2$ is 7. $E(x, y)$ is generated as per the Eq. (18).

$$E(x, y) = FrT_{Z_2}(H_2 * CSPM_2) \tag{18}$$

The schematic diagram of decryption process is shown in Fig. 4.

Step 1: The encrypted image from the encryption process is applied to a Fresnel transform with $-z_2$. It is then multiplied by the conjugate of second chaotic mask ($CSPM_2$). Key1 is

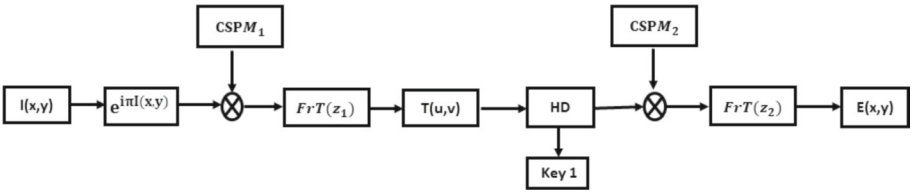


Fig. 3 Schematic diagram—encryption

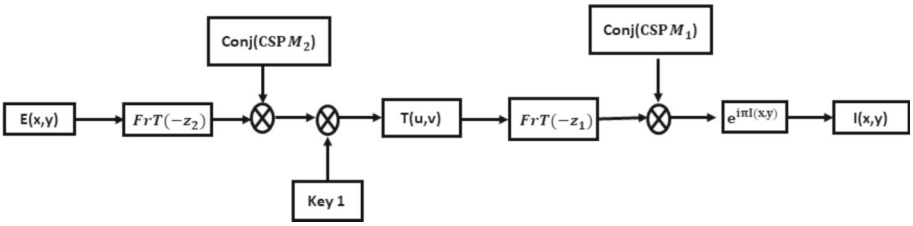


Fig. 4 Schematic diagram—decryption

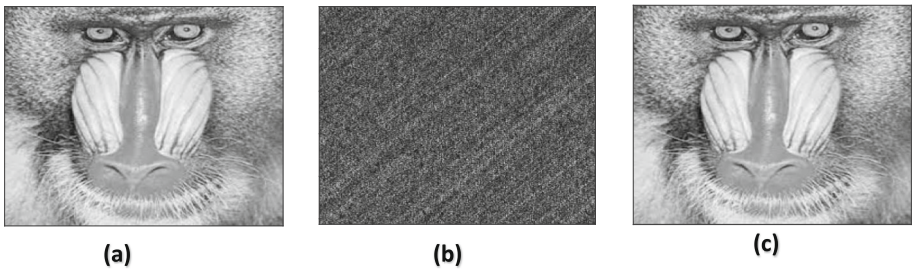


Fig. 5 shown results for grey image **a** Original **b** Encrypted; **c** Decrypted

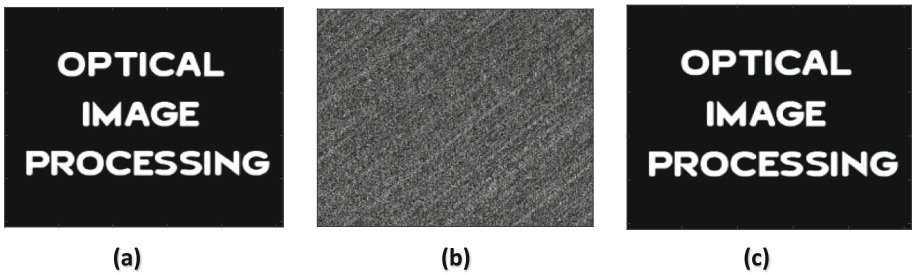


Fig. 6 shown results for binary image **a** Original **b** Encrypted; **c** Decrypted

multiplied with the product, and it is shown in Eq. (18).

$$T(u,\ v) = \left[\left\{\left(FrT_{-z_2}E(x,\ y)\right) * Conj(CSPM_2)\right\} \times Key1\right] \tag{19}$$

Step 2: To recover the image, Fresnel transform is performed on $T(u,\ v)$ and the resultant image is multiplied with conjugate of first chaotic mask. Finally, the angle gives the required decrypted image (Figs. 5 and 6).

$$I(x,\ y) = angle\left[\{(FrT_{-z1}(T(u,\ v)) * Conj(CSPM_1)\}\right] \tag{20}$$

4 Performance analysis (Alexan et al., 2023a, 2023b, 2023c, 2024; Anjana et al., 2023; El-Damak et al., 2024; Gaur et al., 2023; Girija & Singh, 2019; Sachin, 2023; Singh et al., 2023)

4.1 3D plot analysis (Gaur et al., 2023; Sachin, 2023)

The three-dimensional mesh analysis for the proposed system is shown in Fig. 7. As seen in the figure, the three-dimensional mesh of the input image $I(x,\ y)$ is similar to that of decrypted image. The original greyscale and binary images are displayed in Fig. 7a and d, respectively. Figure 7b and e display the encrypted image mesh analysis. Figure 7c and f display the image that has been decoded. From the Fig. 7c and d, mesh plot of encrypted images are looking similar; therefore attacker may not be able to retrieve the plain image information from any encrypted images.

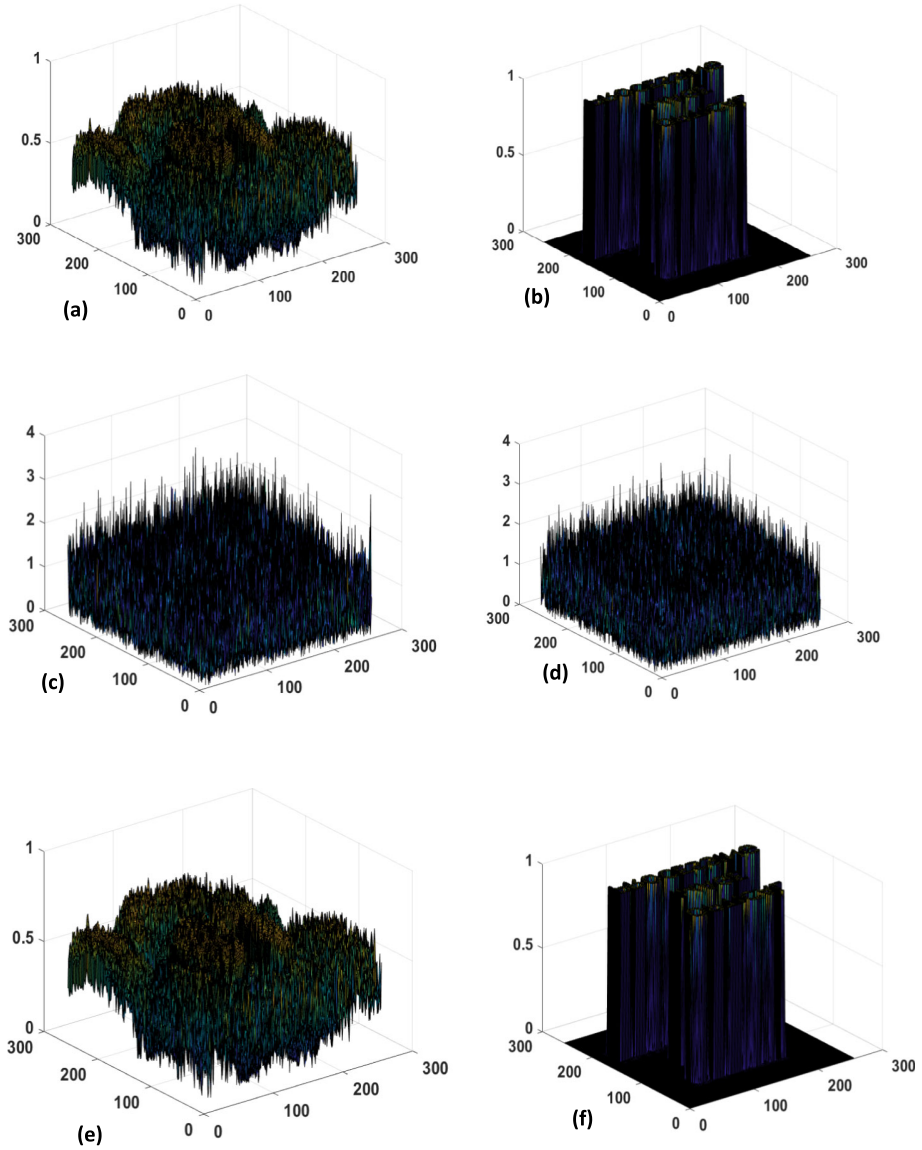


Fig. 7 3D plot for both Grey and binary image

4.2 Mean squared error (MSE)

MSE is a commonly used metric to assess the quality and execution of both encrypted and decrypted images. It represents the average squared difference between the reconstructed and original pixel values. In the context of image encryption, MSE quantifies the extent of damage or loss of accuracy caused by the encryption and decryption operations. A lower MSE value indicates better information security and higher image quality, as it reflects a closer match between the original and reconstructed images. The succeeding formula can be

Table 1 MSE and PSNR values of our proposed system

Image type	Mean square error	Peak signal to noise ratio
Grey image	1.68×10^{-31}	307.74 dB
Binary image	1.62×10^{-31}	307.87 dB

utilized to calculate MSE.

$$MSE = \frac{1}{M \times N} \sum_{i=1}^{M} \sum_{j=1}^{N} \left(I(x,\ y) - I'(x,\ y)\right)^2 \tag{21}$$

where the size of the image is $M \times N$ pixels. The original image is represented by $I(x,\ y)$ and recovered image is denoted by $I'(x,\ y)$. The obtained MSE value for the proposed system is 1.68×10^{-31} for the greyscale image and 1.62×10^{-31} for binary image. It is evident from these MSE values the proposed solution demonstrates superior efficacy and robustness. MSE values for this proposed system has been calculated and it is mentioned in Table 1.

4.3 Peak signal to Noise ratio (PSNR)

Peak signal to Noise ratio (PSNR) is used to evaluate the performance and assess the quality of both decrypted and encrypted images. It measures the correlation between maximum power the original image and the noise level introduced by the encryption and decryption processes. PSNR quantifies the degree of similarity and dependability between the original and restored pictures. A higher PSNR value indicates better quality retention and less distortion. The formula for PSNR is given below:

$$PSNR = 10 \times \log 10\left(\frac{Max^2}{MSE}\right) \tag{22}$$

where Max denotes the image's upper limit pixel value (for example,.255) in 8-bit images. Generally, PSNR is expressed in decibels (db). The PSNR for reconstructed images is 307.74 dB for greyscale images and 307.87 dB for binary images when compared to the original image. The estimated PSNR values for this suggested system are listed in Table 1.

4.4 Key sensitivity analysis (Alexan et al., 2023a, 2023b, 2023c, 2024; Anjana et al., 2023; El-Damak et al., 2024; Gaur et al., 2023; Girija & Singh, 2019; Sachin, 2023; Singh et al., 2023)

Key-sensitive analysis has been conducted to evaluate how changes in encryption and decrypting keys affect the system's performance and security. The proposed system's sensitivity to changes in any critical factor has been assessed by examining its resilience to attacks and the quality of the decrypting images. Figure 8 illustrates the key sensitivity analysis of our proposed system. Figure 8a shows the outcome when the proposed method decrypts data without using an HD key. The decryption procedure without chaotic 1 and chaotic 2 is depicted in Fig. 8b and c. If both chaotic masks 1 and chaotic masks 2 are swapped, the resultant image is denoted in Fig. 8d. This research clarifies that the decryption procedure cannot be carried out if an attacker gains knowledge of the keys, highlighting the strength and security of the

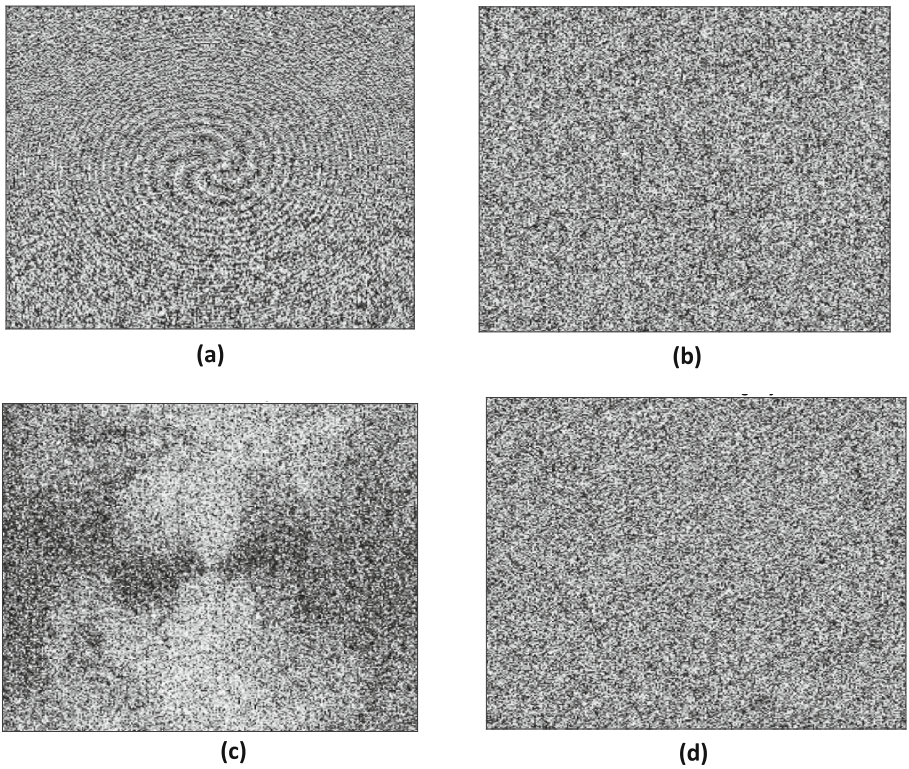


Fig. 8 depicts the decryption for grey image **a** without CSPM1; **b** without CSPM2; **c** when the chaotic masks are swapped with each other; **d** without the Hessenberg key

proposed solution. Figure 9 presents the MSE plot of both grey scale and binary image for Fresnel transforms.

4.5 Occlusion analysis (Alexan et al., 2023a, 2023b, 2023c, 2024; Anjana et al., 2023; El-Damak et al., 2024; Gaur et al., 2023; Girija & Singh, 2019; Sachin, 2023; Singh et al., 2023)

Occlusion analysis has been conducted to assess the robustness and resilience of the proposed strategy against data loss transmission. This attack involves intentionally removing certain areas or segments of the encrypted images. The obscured sections may represent parts of the encrypted image that have been manipulated, physically damaged, or distorted to evaluate the system's resilience. Figure 10 represents the results of occlusion analysis. Figure 10a, b and c show the data hidden at 20%, 45% and 60% respectively. The corresponding decrypted images (both binary and grey scale) are shown in Fig. 10d, g, e, h, f and i. From the Fig. 10, it can be observed that when portions of the encrypted image are lost, it becomes difficult to recover the original image, demonstrating the robustness of proposed system. Figure 11 denotes the occlusion plot for both grey scale image and binary images with the percentage of occlusion.

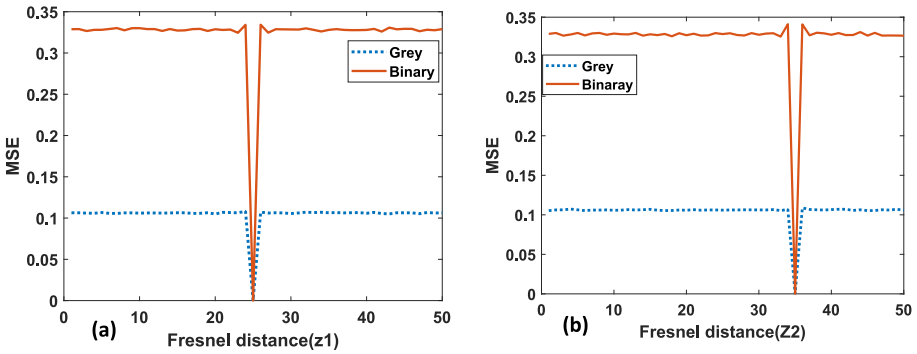


Fig. 9 MSE plot with **a** Fresnel distance z_1; **b** Fresnel distance z_2

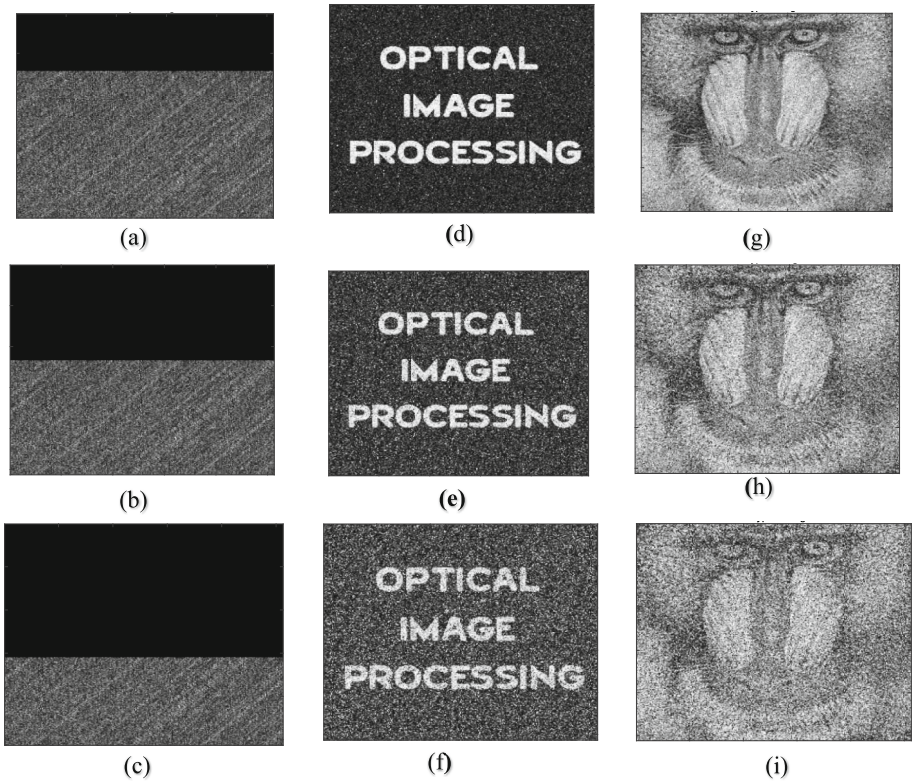


Fig. 10 occlusion results with 20%, 45% and 60%

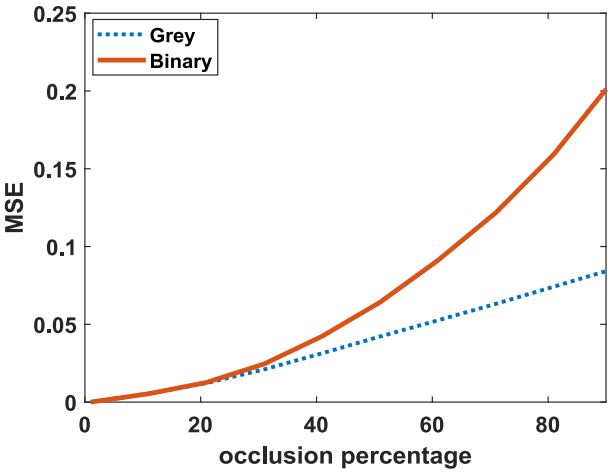


Fig. 11 represents MSE plot with varying percentage of occlusion

4.6 Correlation coefficient analysis (Alexan et al., 2023a, 2023b, 2023c, 2024; Anjana et al., 2023; El-Damak et al., 2024; Gaur et al., 2023; Girija & Singh, 2019; Sachin, 2023; Singh et al., 2023)

The correlation between neighbouring pixels is an important metric for evaluating the effectiveness of an image encryption system. Typically, neighbouring pixels in a plaintext image have a strong correlation with one another, whereas in a ciphertext image, encrypted using a robust encryption algorithm, the correlation between its neighbouring pixels is minimal. For the adjacent pixel correlation simulation, a greyscale image is selected as the plaintext image, and its encrypted counterpart is used as the ciphertext image for comparison. A total of 10,000 pairs are randomly selected from both the plaintext and ciphertext images to compute the correlation coefficient in the horizontal, vertical, and diagonal directions. The correlation can be mathematically defined as follows:

$$Cor(X) = \frac{\sum_{t=1}^{m}(x_i - E(x))(y_i - E(y))}{\sqrt{\sum_{t=1}^{m}(x_i - E(x))^2 \sum_{t=1}^{m}(y_i - E(y))^2}} \tag{21}$$

where, x_i and y_i denotes two randomly selected pixels; m represents the selected pixel pair and $E(.)$ refers to the average calculating function. The correlation results of the encrypted image used in the proposed system, along with the original image's correlation in the horizontal, vertical, and diagonal directions, are shown in Fig. 12. The figure clearly demonstrates that the pixels in the original image have a strong correlation with one another, while the pixels in the encrypted image exhibit a much weaker correlation.

4.7 Noise attack analysis (Alexan et al., 2023a, 2023b, 2023c, 2024; Anjana et al., 2023; El-Damak et al., 2024; Gaur et al., 2023; Girija & Singh, 2019; Sachin, 2023; Singh et al., 2023)

This analysis aims to evaluate the ability of the proposed system to preserve image quality and to recover the original data in the presence of noise. To simulate a noise attack, Gaussian

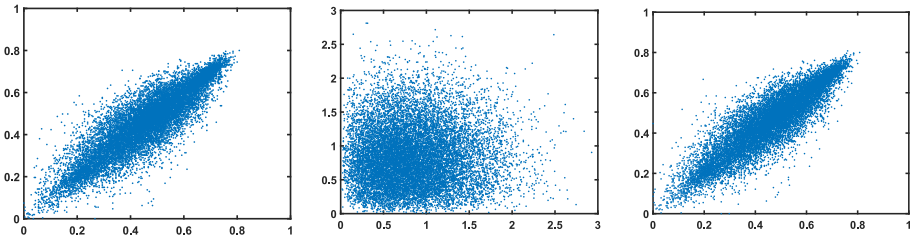


Fig. 12 CC plot for grey image **a** original; **b** encrypted; **c** decrypted

noise with varying variance, such as $k = 0.4$ and 0.9, was applied to the encrypted image. The performance was then assessed by comparing the degree of degradation in the encrypted images with that of the original image. Figure 13a shows the decrypted image with noise strength as 0.4 and Fig. 13b represents the decrypted image with the noise strength 0.9. Figure 14 represents the CC plot with varying noise strength.

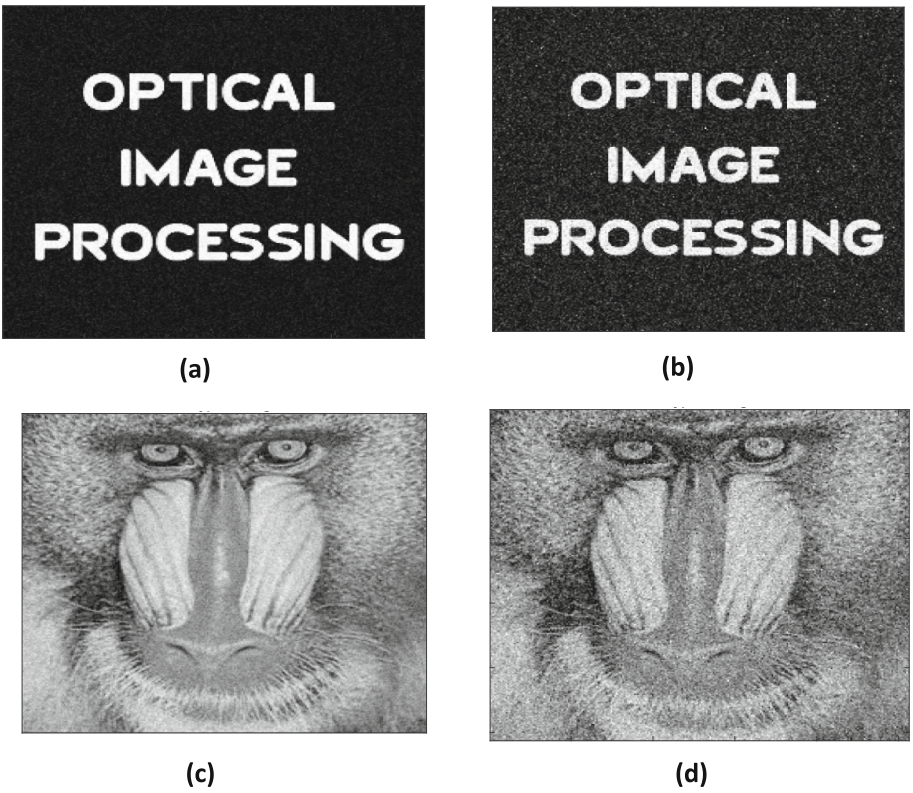


Fig. 13 Decrypted image with noise strength for k = 0.4 and k = 0.9

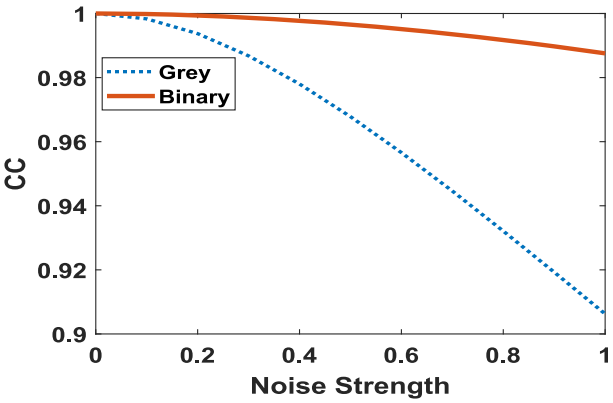


Fig. 14 Noise strength versus CC plot

5 Conclusion

This proposed system introduces a phase encryption algorithm utilizing chaotic structured phase masks and the Hessenberg Decomposition (HD). The system is categorised as an asymmetric cryptosystem because HD provides both asymmetry and non-linearity. The key components of chaotic structured phase masks and Fresnel transform parameters are used to contribute the security of the system. To assess the robustness and security of the system, various analyses were conducted, including MSE, PSNR, noise analysis, occlusion analysis and key sensitivity analysis. The results of these analyses demonstrate that the hybrid cryptosystem is highly secure and can be applied in scenarios where both asymmetric and symmetric encryptions are required. As Hessenberg decomposition is used to generate the key of the proposed crytposystem and the limitation of using HD is, it may become more computationally complex when working with larger size of image. Since, our proposed system also uses chaotic maps, it could be a time- consuming process or resource-heavy for real time applications.

Author contributions R.G and K.S drafted the manuscript; K.S. performed Simulation; R.G, K.S and H.S reviewed the manuscript;

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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